

## CONTROL STATEMENTS:-

### “If” – statements:

1. WAP to check if a given number is positive.

The screenshot shows the SIMATS Online Programming Lab interface. The header is dark blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING" in a yellow box. A user profile "Ashira Bridget Susan S" with ID "192524316RD" is in the top right. The main area is divided into "PROGRAM EDITOR" and "INPUT". The "PROGRAM EDITOR" has a "Python" dropdown, "Run" and "Save" buttons, and a code editor with the following code:

```
1 a = int(input())
2 if a>0:
3     print("Given number is positive")
```

The "INPUT" field on the right contains the number "5".

### OUTPUT

Given number is positive

2. WAP to provide flat Rs. 500 if the purchase amount is greater than 2000.

The screenshot shows the SIMATS Online Programming Lab interface. The header is dark blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING" in a yellow box. A user profile "Ashira Bridget Susan S" with ID "192524316RD" is in the top right. The main area is divided into "PROGRAM EDITOR" and "INPUT". The "PROGRAM EDITOR" has a "Python" dropdown, a "✓ Saved" status, and "Run" and "Save" buttons. The code editor contains:

```
1 a = int(input())
2 if a>2000:
3     print(a-500)
```

The "INPUT" field on the right contains the number "3000".

### OUTPUT

2500

3. WAP to provide bonus mark if the category is sports.

The screenshot shows the SIMATS Online Programming Lab interface. The header is dark blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING" in a yellow box. A user profile "Ashira Bridget Susan S" with ID "192524316RD" is in the top right. The main area is divided into "PROGRAM EDITOR" and "INPUT". The "PROGRAM EDITOR" has a "Python" dropdown, "Run" and "Save" buttons, and a code editor with the following code:

```
1 category = input()
2 if category == "sports":
3     print("Bonus marks provided")
```

The "INPUT" field on the right contains the word "sports".

## OUTPUT

```
Bonus marks provided
```

4. WAP to check if the number is Zero.

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PYTHON PROGRAMMING

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192524316RD

PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a==0:
3     print("Zero")
```

INPUT

0

## OUTPUT

```
Zero
```

## “If-Else” – Statement:-

5. WAP to check whether the given years is a leap year or not.

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PROGRAM EDITOR

Python ✓ Saved Run Save

```
1 year = int(input())
2 if (year%4==0 and year%100!=0) or (year%400==0):
3     print(year, " is a leap year")
4 else:
5     print(year, " is not a leap year")
```

INPUT

2024

OUTPUT

2024 is a leap year

6. WAP to check the eligibility for voting.

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PROGRAM EDITOR

Python Run Save

```
1 age = int(input())
2 if age>=18:
3     print("Eligible for voting")
4 else:
5     print("Not eligible for voting")
```

INPUT

27

OUTPUT

Eligible for voting

## 7. WAP to check if even/odd

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if(a%2==0):
3     print("It is even")
4 else:
5     print("It is odd")
```

INPUT

6

OUTPUT

It is even

## 8. WAP to check if the given is a 2 digit number or not.

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a>9 and a<100:
3     print("It is a 2-digit number")
4 else:
5     print("It is not a 2-digit number")
```

INPUT

25

OUTPUT

It is a 2-digit number

9. WAP to check if a number is divisible by 3 and 5 or not.

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a%3==0 and a%5==0:
3     print(a, "is divisible by 3 or 5")
4 else:
5     print(a, "is not divisible by 3 or 5")
```

INPUT

15

OUTPUT

15 is divisible by 3 or 5

10. WAP to check if a given number is divisible by 5 or not, if not, print the remainder.

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a%5==0:
3     print(a, "is divisible by 5")
4 else:
5     print(a, "is not divisible by 5 and the remainder is ", (a//5))
```

INPUT

6

OUTPUT

6 is not divisible by 5 and the remainder is 1

11. WAP to see if a number is positive/negative.

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python ▼

Run

Save

```
1 a = int(input())
2 if(a>0):
3     print("It is positive")
4 else:
5     print("It is negative")
```

INPUT

6

OUTPUT

It is positive

## “If-Elif-Else” – statements:-

12. WAP to find the largest of given 3 numbers.

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 b = int(input())
3 c = int(input())
4
5 if(a>b) and (a>c):
6     largest = a
7 elif b>c:
8     largest = b
9 else:
10    largest = c
11 print(largest, " is the largest of theh given 3 number.")
```

INPUT

5  
9  
4

### OUTPUT

```
9  is the largest of theh given 3 number.
```

13. WAP to check the given number is positive or negative or zero.

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a>0:
3     print(a, " is positive")
4 elif a==0:
5     print("Zero")
6 else:
7     print(a, " is negative")
```

INPUT

-3

### OUTPUT

```
-3  is negative
```

14. WAP to check if a number is divisible by 5 and 10 or by 5 or 10.

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a%5==0 and a%10==0:
3     print(a, "is divisible by 5 and 10")
4 elif a%5==0 or a%10==0:
5     print(a, "is divisible by 5 or 10")
6 else:
7     print(a, "is not divisible 5 or 10")
```

INPUT

50

OUTPUT

50 is divisible by 5 and 10

15. WAP to see if a given number is single/double/above 2/negative no.

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a>=0 and a<=9:
3     print("It is a single digit")
4 elif a>9 and a<100:
5     print("It is a double digit")
6 elif a>2:
7     print("It is above 2")
8 else:
9     print("It is a negative number")
```

INPUT

-9

OUTPUT

It is a negative number



16. WAP to check the age category.

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PROGRAM EDITOR

Python

Run

Save

```
1 age = int(input())
2 if age < 12:
3     print("You are a child!")
4 elif age >12 and age<=19:
5     print("You are a teenager")
6 elif age>19 and age<=35:
7     print("You are a young adult")
8 else:
9     print("You are a senior citizen")
```

INPUT

66

OUTPUT

You are a senior citizen

17. WAP for a student Mark System

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input())
2 if a>= 90:
3     print("S Grade")
4 elif a>=80:
5     print("A Grade")
6 elif a>=70:
7     print("B Grade")
8 elif a>=60:
9     print("C Grade")
10 elif a>=50:
11     print("D Grade")
12 else:
13     print("F Grade")
```

INPUT

95

OUTPUT

S Grade

## 18. WAP for a Traffic Light System

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PROGRAM EDITOR

Python

Run

Save

```
1 color = input()
2 if color == "red":
3     print("STOP!")
4 elif color == "yellow":
5     print("SLOW DOWN/GET READY!")
6 elif color == "green":
7     print("GO!")
8 else:
9     print("INVALID COLOR!")
```

INPUT

blue

### OUTPUT

INVALID COLOR!

## 19. WAP to find the Roots of Quadratic Equation

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PROGRAM EDITOR

Python

Run

Save

```
1 a = float(input())
2 b = float(input())
3 c = float(input())
4
5 d = (b**2) - (4*a*c)
6
7 if d>0:
8     r1 = (-b + (d**0.5)) / (2*a)
9     r2 = (-b - (d**0.5)) / (2*a)
10    print("Two real roots: ", r1, "and", r2)
11 elif d==0:
12     r = -b / (2*a)
13     print("One real root:", r)
14 else:
15     print("The roots are imaginary/complex")
```

INPUT

1  
-5  
6

### OUTPUT

Two real roots: 3.0 and 2.0

## “While” – Loop Statements:-

### 1. Display 1 to 10

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PROGRAM EDITOR

Python ▼

Run

Save

```
1 i = 1
2 while(i<=10):
3     print(i)
4     i+=1
```

INPUT

Your INPUT goes here!

OUTPUT

1  
2  
3  
4  
5  
6  
7  
8  
9  
10

### 2. Display 10 to 1

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PROGRAM EDITOR

Python ▼

Run

Save

```
1 i = 10
2 while(i>=1):
3     print(i, end=" ")
4     i-=1
```

INPUT

Your INPUT goes here!

## OUTPUT

```
10 9 8 7 6 5 4 3 2 1
```

3. Display odd numbers from 1 to 20

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python ▼

Run Save

```
1 num = 1
2 while(num <=20):
3     if(n%2!=0):
4         print(num, end = " ")
5         num+=1
```

INPUT

5

4. WAPP to find sum of n numbers.

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python ▼

Run Save

```
1 i = 0
2 sum = 0
3 while (i <=5):
4     sum = sum + i
5     i = i + 1
6 print("Sum of series: ", sum)
```

INPUT

Your INPUT goes here!

## OUTPUT

```
Sum of series: 15
```

5. WAPP to find the sum of even numbers.

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PYTHON PROGRAMMING

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192524316RD

PROGRAM EDITOR

Python

Run

Save

```
1 i=0
2 sum = 0
3 while (i <= 5):
4     if(i%2==0):
5         sum = sum+i
6     i = i+1
7 print(sum)
```

INPUT

10

OUTPUT

6

6. Write a Python Program to find sum of digits of a given number

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PYTHON PROGRAMMING

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192524316RD

PROGRAM EDITOR

Python

Run

Save

```
1 n = int(input())
2 org = n
3 sum = 0
4 while (n>0):
5     a = n % 10
6     sum +=a
7     n = n//10
8 print("Sum of the digits of", org , "is", sum)
```

INPUT

234

OUTPUT

Sum of the digits of 234 is 9

7. WAPP to print the Fibonacci Series.

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python

Run

Save

```
1 n = int(input())
2 a = 0
3 b = 1
4 count = 0
5 print("Fibonacci Series: ")
6 while(count<n):
7     print(a, end = " ")
8     c = a + b
9     a = b
10    b = c
11    count += 1
```

INPUT

10

OUTPUT

Fibonacci Series:  
0 1 1 2 3 5 8 13 21 34

8. WAPP to reverse the given number.

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PROGRAM EDITOR

Python

Run

Save

```
1 n = int(input())
2 rev = 0
3 while(n>0):
4     a=n%10
5     rev=rev*10+a
6     n=n//10
7 print(rev)
```

INPUT

12345

OUTPUT

54321

9. WAPP to find whether the number is armstrong or not.

## PROGRAM EDITOR

Python ▼

Run

Save

## INPUT

153

```
1 n = int(input())
2 org=n
3 sum = 0
4 while(n>0):
5     a=n%10
6     sum=sum+(a**3)
7     n=n//10
8 print(sum)
9
10 if sum==org:
11     print("It is an Armstrong Number")
12 else:
13     print("It is not an Armstrong Number")
```

## OUTPUT

```
153
It is an Armstrong Number
```

10. Check number is palindrome or not.

## PROGRAM EDITOR

Python ▼

Run

Save

## INPUT

12521

```
1 n = int(input())
2 org=n
3 rev = 0
4 while(n>0):
5     a=n%10
6     rev=rev*10+a
7     n=n//10
8 print(rev)
9
10 if rev==org:
11     print("It is a Palindrome")
12 else:
13     print("It is not a Palindrome")
```

```
12521
It is a Palindrome
```